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Management of head injuries

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Objectives

- 1. Management of different types of head injury.
- 2. Monitoring of patients.
- 3. Management of head injury complications.



Initial management of head injury

The key aspects in the management of patients following head injury involve:

- accurate clinical assessment of the neurological and other injuries
- determination of the pathological process involved the concept that a change in the neurological signs indicates a progression or change in the pathological processes.
- It is essential to avoid hypoxia and hypotension as these will cause further brain injury

Pathophysiology of head injury

The management of head injury has been based on the concept of **primary** and **secondary** brain injury.

- The primary brain injury was defined as the irreversible pathology sustained at the time of the trauma,
- whereas the secondary brain injury has been considered the subsequent or progressive brain damage that occurs due to an evolving pathology following the injury.



Management of Traumatic Head Injury

- Maximize oxygenation and ventilation
- Support circulation / maximize cerebral perfusion pressure

$$\text{CPP} = \text{MAP} - \text{ICP} \text{ (if ICP is higher than JVP)}$$

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Cerebral perfusion pressure: is the net pressure gradient causing cerebral blood flow to the brain.

$$\text{Mean arterial pressure: } \underline{\text{SBP} + 2(\text{DBP})}$$



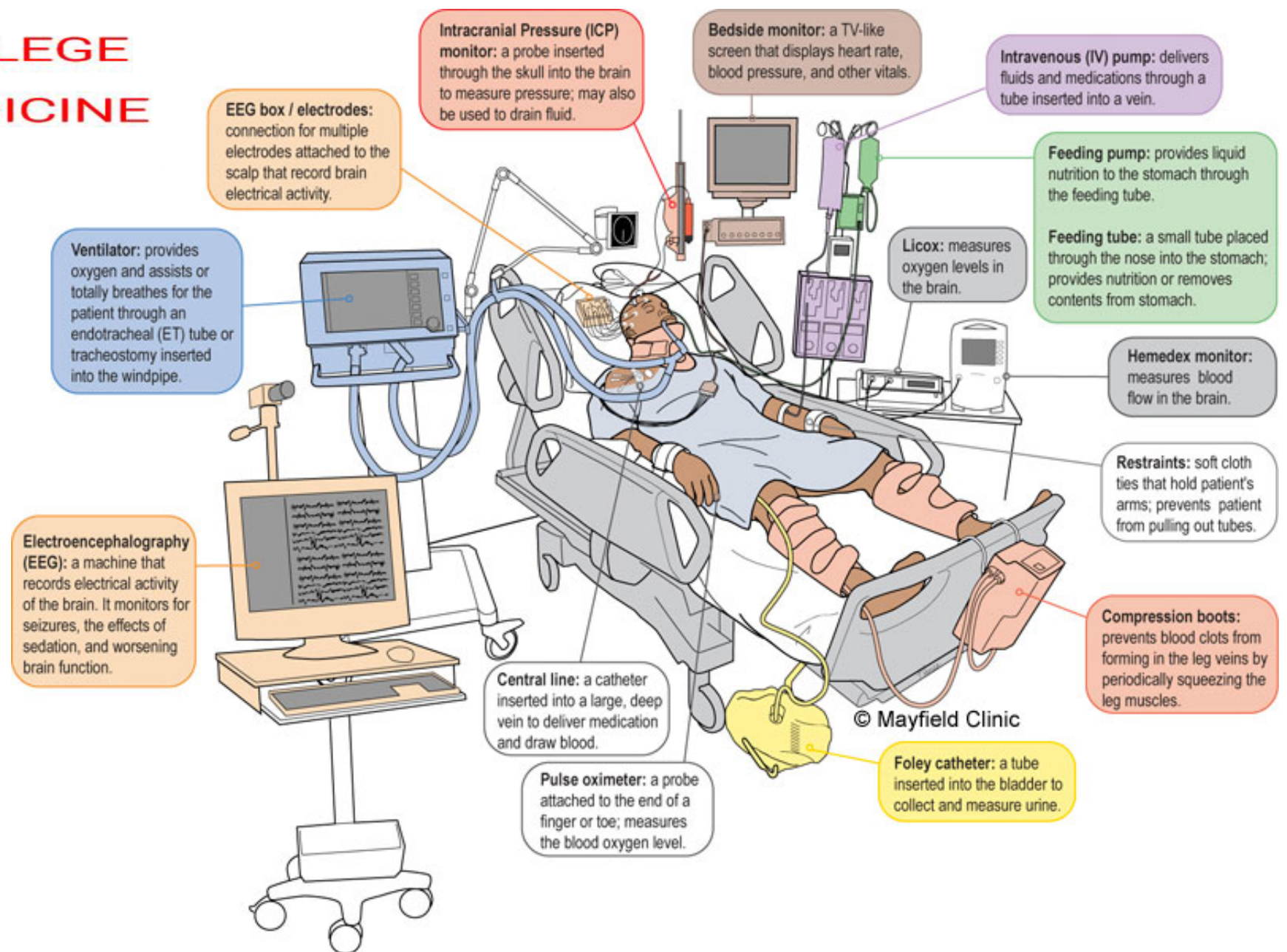
With normal conditions of MABPs of 80 to 100 mm Hg and ICPs of 5 to 10 mm Hg, CPP can be expected to be 70 to 85 mm Hg.

With conventional limits of intact autoregulation, CBF remains normal as CPP varies between 60 and 140 mm Hg.

- Decrease intracranial pressure
- Decrease cerebral metabolic rate

Monitoring

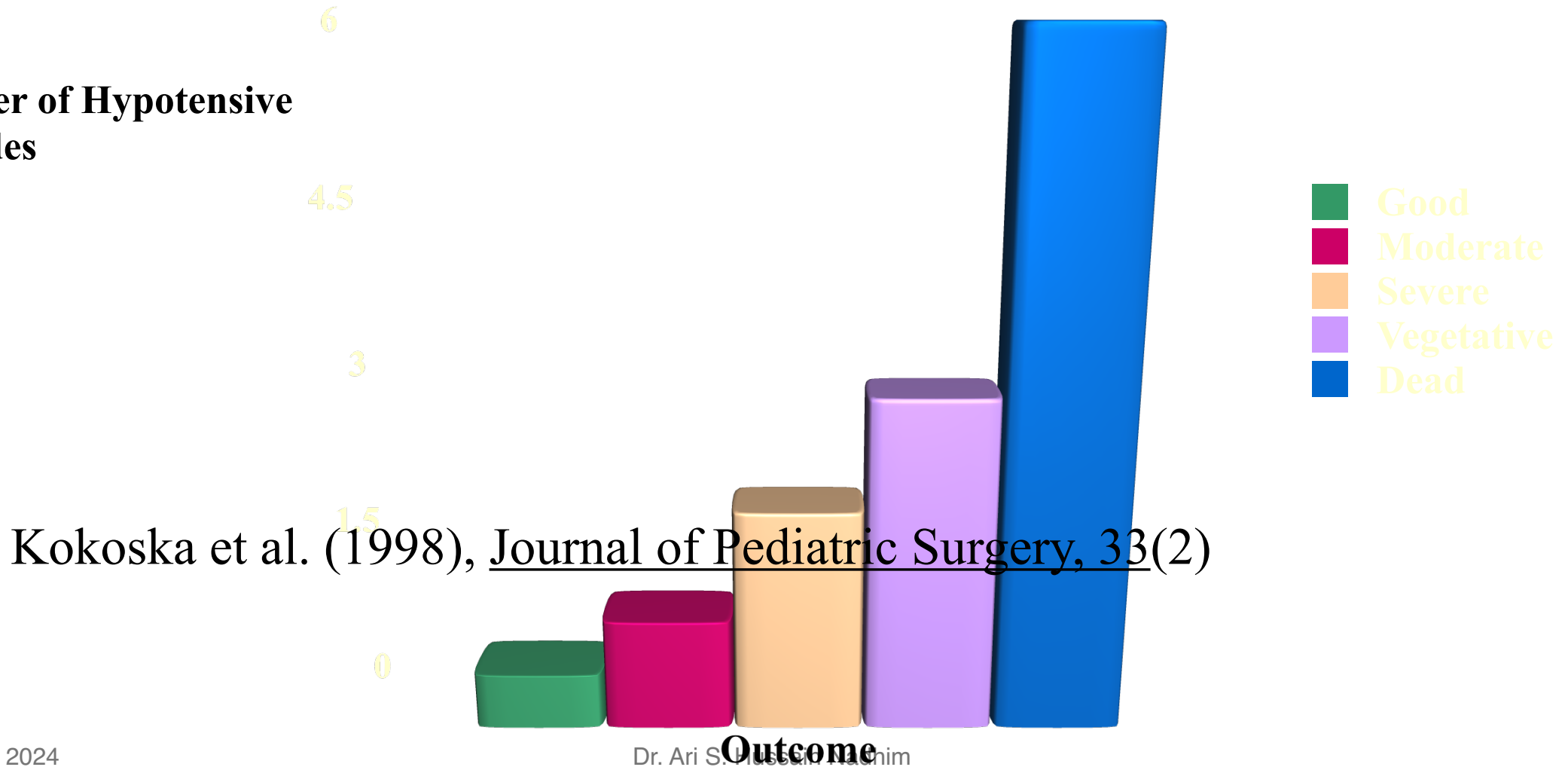
- Serial neurologic examinations
- Circulation / Respiration
- Intracranial Pressure
- Radiologic Studies
- Laboratory Studies





Circulatory Support: Maintain Cerebral Perfusion Pressure

**Number of Hypotensive
Episodes**





- Reduce edema
- Promote venous return
- Reduce cerebral metabolic rate
- Reduce activity associated with elevated ICP



- For a **mild head injury**, no specific treatment may be needed. However, closely watch the person for any concerning symptoms over the next 24 hours.
- The symptoms of a **serious head injury** can be delayed. While the person is sleeping, wake him or her every 2 to 3 hours and ask simple questions to check alertness



Indications for admission to hospital:

- **Loss of consciousness.**
- **Persistent drowsiness.**
- **Focal neurological deficit.**
- **Skull fracture.**
- **Persisting nausea & vomiting**
- **Elderly & infant.**
- **W.**



Minor head injury

The following are the minimal criteria for **obligatory CT scan and admission to hospital:**

- loss of consciousness (post-traumatic amnesia) of greater than 10 minutes
- persistent drowsiness
- focal neurological deficits
- skull fracture
- persisting nausea or vomiting after 4 hours' observation
- intracranial pathology noted on a CT scan
- if the patient does not have adequate care at home



- Signs of deterioration:
 - **Becomes unusually drowsy**
 - **Develops a severe headache or stiff neck**
 - **Vomits more than once**
 - **Loses consciousness (even if brief)**
 - **Behaves abnormally**



indications that a head injury is more serious than a concussion and requires emergency treatment include:

- **Changes in size of pupils**
- **Clear or bloody fluid draining from the nose, mouth, or ears**
- **Convulsions**
- **Distorted facial features**
- **Drop in blood pressure**
- **Facial bruising**
- **Fracture in the skull or face**
- **Impaired hearing, smell, taste, or vision**
- **Inability to move one or more limbs**
- **Irritability**
- **Loss of consciousness**
- **Low breathing rate**
- **Restlessness, clumsiness, or lack of coordination**
- **Severe headache**
- **Slurred speech or blurred vision**
- **Stiff neck or vomiting**
- **Sudden worsening of symptoms after initial improvement**
- **Swelling at the site of the injury**
- **Persistent vomiting**

Management

1) Supportive Measures:

- Endotracheal intubation for patients with decreased level of consciousness and poor airway protection.
- Cautiously lower blood pressure to a MAP less than 130 mm Hg, but avoid excessive hypotension.^[10]
- Rapidly stabilize vital signs, and simultaneously acquire emergent CT scan.
- Maintain euvolemia, using normotonic rather than hypotonic fluids, to maintain brain perfusion without exacerbating brain edema
- Avoid hyperthermia.
- Facilitate transfer to the operating room or ICU.



Management

2) Decrease cerebral edema:

- Modest passive hyperventilation to reduce PaCO₂
- Mannitol, 0.5-1.0 gm/kg slow iv push
- Furosemide 5-20 mg iv
- Elevate head 20-30 degrees, avoid any neck vein compression
- Sedate and paralyze if necessary with morphine and vecuronium (struggling, coughing etc will elevate intracranial pressure)



Management

4) Medical therapy:

- Antihypertensives - reduce blood pressure to prevent exacerbation of intracerebral hemorrhage in hypertensive encephalopathy. Eg Nicardipine, labetolol; CCB help relieve vasospasm in SAH and decrease further damage
- Diuretics - Mannitol, CAI
- Anticonvulsants – reduce frequency of seizures and prophylaxis of seizures eg: Fosphenytoin
- Antipyretics- to Rx fever and pain relief eg: Acetaminophene
- Antidote-
 - VitK/FFP for warfarin overdose;*
 - protamine for heparin overdose*
- Antacids- prophylaxis for Cushing's gastric ulcer eg: Famotidin
- Glucocorticoids may help reduce the head and neck ache caused by the irritative effect of the subarachnoid blood.



- **If there is no surgical lesion, or following the operation:**
 - **Observation and GCS chart**
 - **Decrease intracranial brain swelling**
 - **Airway management**
 - **Elevation of the head of the bed 20°**
 - **Fluid and electrolyte balance**
 - **Blood replacement with colloid or blood and not crystalloid**
 - **No steroids**



– **Management of conditions resulted from head injury**

- **Severe hyponatraemia due to excessive fluid intake or inappropriate excessive secretion of ADH**
- **Hypernatraemia due to inadequate fluid intake.**
- **Diabetes insipidus**

The techniques used to control intracranial pressure are as follows:

- Controlled ventilation, maintaining PaCO₂ at 30–35 mmHg.
- Reduction of the PaCO₂ will reduce cerebral vasodilatation and consequently decrease the intracranial pressure.
- If the pressure remains elevated despite hyperventilation CSF can be drained from a ventricular catheter if this has been inserted.
- Diuretic therapy utilizing intermittent administration of mannitol or frusemide (furosemide) can be used if the preceding techniques have failed to control the intracranial pressure.
- Mannitol is an osmotic diuretic and may also exert its effect by increasing serum osmolality and drawing water out of the brain.
- The usual dose is 0.5–1.0 g/kg.
- The serum osmolality should not exceed 320 mosmol/kg.
- Barbiturate therapy



Steroids.

- Although steroids dramatically reduce the oedema around cerebral tumours they have little effect in controlling the brain swelling following a head injury.
- possible complications, such as gastrointestinal bleeding, poor wound healing and infection, may result from their administration.
- Hypothermia.
- Hyperbaric oxygen
- Decompressive craniotomy

Cerebral herniation

- **Supratentorial**
 - **Uncal**
 - **Central (transtentorial)**
 - **Cingulate (Subfalcine)**
 - **Transcalvarial**
- **Infratentorial**
 - **Upward cerebellar**
 - **Tonsillar (foramen magnum)**

